Proposed manuscript changes — DRAFT

Manuscript 2067810, revision 1. **For the author's approval: nothing below is applied yet.** Kind: T = text only, can be written now; R = needs the results of cohort B2; T+R = both. Page effect: − shorter, 0 neutral, + longer.

Part A. Length: as short as possible

The journal sets no length limit (writing rules, last updated 2025-02-22); published articles run 7–17 pages. Proposal: **a hard ceiling of 14 pages** (the length submitted), aiming for 12–13, with every table and paragraph earning its place. The page budget in `build_revision.py` goes from 17 to 14 once approved.

What the 14 pages hold now (body 5,500 words; 9 tables, 2 figures, 4 equations, 38 references):

Current section	Words	Proposal
Introduction	493	Rewrite, 400 (P-01)
Background (LIME, SHAP, evaluation)	424	Condense to 300, plus 150 on recent comparisons (P-02)
Cohort and RQs	223	Replace, 150 (P-04, P-05)
Design / data / explainers	293	Replace with Table 1 (design) and 250 words (P-06–P-08)
Metrics	421	Keep, add conventions, 380 (P-09)
Inference	64	Rewrite, 220 (P-11)
Reproducibility / infrastructure	97	Rewrite, 120 (P-10)
Context: earlier results	166	**Delete** (P-03)
Primary paired inference (Table 1, Fig. 1)	336	Replace: four-explainer table, 400 (P-12)
Seed reproducibility (Table 2)	206	**Delete**; seed variance enters the hierarchical analysis (P-13)
Runtime (Table 3, Fig. 2)	382	Replace: cost table with wall time, CPU time and model calls, 380 (P-14, P-15)
Cross-dataset (Tables 4, 5)	444	One table, all endpoints, 280 (P-17)
Kernel width (Table 6)	339	Replace with the sweep: one multi-panel figure, 330 (P-16)
Masking (Table 7)	294	Merge into a "validity of the measurements" section, 300 with stability (P-18)

Current section	Words	Proposal
Discussion (three parts, Tables 8, 9)	923	650; one compact table (P-19, P-20)
Conclusions	210	180 (P-21)
Statements, data availability	132	130
Total body	**5,500**	**≈ 5,050**; tables 9 → 5; figures 2 → 2

Proposed outline (target pages in the journal template):

1. Introduction (1.0)
2. Background and related work (1.0)
3. Materials and methods (3.0): 3.1 questions and design (Table 1) · 3.2 data and models · 3.3 explainers and configurations · 3.4 metrics · 3.5 statistical analysis · 3.6 cost measurement and host
4. Results (4.5): 4.1 main benchmark (Table 2) · 4.2 cost (Table 3) · 4.3 configuration sensitivity (Fig. 1) · 4.4 validity of the stability and fidelity measurements · 4.5 cross-dataset results (Table 4)
5. Discussion (1.5): what holds and under which conditions (Table 5) · limitations
6. Conclusions (0.4)
7. References (≈ 2.0); statements (0.3)

Detail that does not fit (per-dataset tables, hyperparameters, sweep grids, conventions): a **supplementary file**, if the author approves (D-B), instead of an appendix that adds pages.

Part B. Proposed manuscript changes

Kind: **T** = text that can be written now; **R** = needs cohort B2 results; **T+R** = method text now, numbers later. Page effect: − shorter, + longer, 0 neutral.

ID	Kind	Items	Where (proposed §)	Change	Pages
P-01	T	R1-1, R1-M1, R2-8, R1-8	§1	New introduction around a practitioner's decision (which explainer, for which model, under which budget). It states the gap: comparisons with matched conditions and configuration-aware results are scarce. Three contributions are stated without reference to the earlier study:	–

ID	Kind	Items	Where (proposed §)	Change	Pages
				(1) a self-contained paired benchmark of four explainers on one host with hardware-independent cost; (2) configuration-aware results; (3) conditional recommendations	
P-02	T	R1-9	§2	Condense the LIME/SHAP background. Add a paragraph on recent (2024–2026) empirical comparisons and benchmarks, and position the paper against them. References follow the author rule (verified DOIs; ≥ 80% from 2024 on)	0
P-03	T	R2-6, R1-M1, R2-7	old §4.1	Delete “Context: Earlier Results on This Cohort” and every pointer that sends the reader to [10] for a number (Tables 1 and 5 captions, runtime text)	–
P-04	T	R2-7	§3, everywhere	Masking: remove the author names at old line 184 (“Herrera-Vásquez and Herrero-Uceda [10]”) and the framing “the execution cohort released with an earlier study”. [10] is cited only for the metric definitions, like any other source	–
P-05	T	R1-m2, R2-4	§3.1	Research questions with one notation: paired difference Δ = SHAP − X for X ∈ {LIME, Anchors, DiCE}; directional hypotheses H1, H3, H4. **H2 (sparsity) becomes descriptive**, with the reason (LIME’s feature budget fixes it)	0
P-06	T	R1-m3, R1-M2	§3.1, Table 1	New design table: datasets, models, explainers and variants, seeds, instances per stratum, stability subset, number of paired cells, timing subset	+
P-07	T	R1-m4, R1-m5, R1-M2	§3.2	Data and models: one fixed split per dataset (seed 42), shared by preprocessing, training and every seed. Stratified sampling of TP/TN/FP/FN instances from the test split only, realized counts; seeds may share instances in the smallest strata; the	0

ID	Kind	Items	Where (proposed §)	Change	Pages
				overlap is measured and reported. Hyperparameters in the supplement	
P-08	T	ED-1, R1-4, R1-M4, R2-3, R1-m12	§3.3	Explainers: SHAP (TreeExplainer for tree models, KernelExplainer otherwise), LIME at the library default kernel width $0.75\sqrt{M}$ (≈ 7.79 on Adult), with and without the 10-feature budget; Anchors (precision 0.95); DiCE (one counterfactual, attribution $\|x - x'\|$). Why each setting, and why Anchors and DiCE are now in the main benchmark	+
P-09	T	R1-m6, R1-m7, R1-3	§3.4	Metric conventions stated: all-zero attribution vectors (counted and reported; stability with and without them); constant vectors in the fidelity correlation (set to 0, and counted); absolute values in Eq. (3). **Correction: Adult has 108 transformed features, not 103**	0
P-10	T	R2-1a, R1-5, R1-M8, R1-m14	§3.6	Cost measurement: one recorded host (CPU, RAM, OS, software versions); a single process, one thread, pinned to one core; warm-up excluded; three repetitions; explainers interleaved per instance; a quiet-host condition. Model calls counted. Parallelism of the quality pass stated; scripts and artifacts described (addresses withheld for review)	+
P-11	T	R1-2, R1-M2, R1-M3, R1-7, R1-M14	§3.5	Statistical analysis that accounts for the hierarchy: a linear mixed model with random intercepts for model and model×seed; a cluster bootstrap over models; Hodges–Lehmann differences and matched rank-biserial correlations next to d_z; Holm correction over the family (endpoints × contrasts); Friedman/Nemenyi for the four explainers. The cell-level test is kept as	+

ID	Kind	Items	Where (proposed §)	Change	Pages
				descriptive	
P-12	R	ED-1, R1-7, R1-M14, R2-6, R1-m8	§4.1, Table 2	Main results for the four explainers: per-method levels and paired differences with CI, p_Holm, d_z, HL and r_rb, in one table with defined columns. The distribution of the paired differences in the supplement (or a small panel)	0
P-13	T	R1-m9	old §4.3	Delete the CV table and its “Reproducible? Yes” labels; seed variability is reported as the variance component of the mixed model	–
P-14	R	R2-1a, R2-1b, R1-5, R1-M8	§4.2, Table 3	Cost: median wall time and CPU time per explanation on one host, and the **number of model calls** and rows evaluated. Theoretical complexity per explainer in one column. FLOPs for logistic regression and MLP only, with the reason	0
P-15	R	R1-M9	§4.2	RF vs XGBoost reversal: the controlled run (matched number of trees and depth, cost against total leaves, one host). Stated as architecture or implementation according to the result [R]	+
P-16	R	R2-2, R1-M5, R1-M11, R1-M12, R2-3, R1-M4	§4.3, Fig. 1	Configuration sensitivity: LIME kernel width × samples, KernelSHAP samples × background, KernelSHAP also on the tree models. Pareto fronts (stability, fidelity, cost) per model in one multi-panel figure, over 3 seeds and all models, on the common set of valid instances. The share of degenerate explanations is reported. Whether the sign of each difference holds over the grid [R]	+
P-17	R	R1-6, R1-M10, R2-5, R1-m11, R1-m12	§4.5, Table 4	Cross-dataset: 7 datasets (Adult and 6 others in a 2×2 of dimension × encoding), all models, four explainers,	0

ID	Kind	Items	Where (proposed §)	Change	Pages
				SHAP stability included, mean ± SD over seeds	
P-18	R	R1-3, R1-M6, R1-M7, R1-M13, R1-m13	§4.4	Validity of the measurements: (a) no-noise control (explainer's own variance); (b) perturbations that keep one-hot features valid; (c) rank and top-k stability on full vectors; (d) zero vectors; (e) the four masking schemes on all models, with the grouping rule stated. Replaces old §4.6 and §4.7	–
P-19	R	R1-M15, R1-10, R1-m15	§5, Table 5	One table "what holds, where, under which configuration", with each claim labelled **confirmatory** (main paired benchmark) or **exploratory** (sweep, controls, cross-dataset). Recommendations restricted to the models, datasets, configurations and host tested	–
P-20	T+R	R1-1, R1-M1, R2-6	§5	One paragraph on the relation to earlier work, in the third person: what was known, what this study adds. No reuse of earlier numbers	0
P-21	R	R1-10, R1-m15	§6	Conclusions that restate only results of Table 2 and Table 5, each traceable	–
P-22	T+R	R1-m1, R1-m15	Abstract	Rewrite: primary benchmark first, then supplementary results, each labelled; ≤ 200 words (journal: 150–200)	0
P-23	T	R1-m10	Figures	Regenerate every figure from committed scripts: labelled axes, units, scales stated, 95% CI defined in the caption, readable at print size	0
P-24	T	R2-7	Build	The masking check also covers the comparison file and the Word metadata (done in the build); the data-availability statement keeps the addresses withheld	0

Part D. Decisions for the author

ID	Decision	Recommendation
D-A	Disclose, in the response (R2-6, R1-M1) and in one sentence of §3, that the earlier cohort was replaced, and why: the train/test overlap in most cells, the lost RF model, the mixed-host timings (`WP0_FINDINGS.md` F7, F8)	**Yes, briefly and factually.** The revision replaces every number, so a reviewer will ask why. Honest disclosure protects the paper (and is consistent with RCA-004)
D-B	A supplementary file for detail tables (hyperparameters, per-dataset results, sweep grids, paired-difference plots) to keep the paper short	**Yes**, uploaded as an additional file. The journal's pages do not mention supplements, so ask the editorial office or check the upload step; fallback: a short appendix
D-C	Page ceiling 14 (the submitted length), aim 12–13	**Yes**; the build warns above it
D-D	Who writes the prose: the assistant drafts each approved change in the manuscript for the author to edit, or prepares briefs	Author's choice
D-E	Approve P-01 to P-24 (all, or one by one). The T items can start now; the R items wait for cohort B2	Approve the T items first: P-01–P-11, P-13, P-23, P-24